



End to End Simulated 5G Network/Test Bed using Open5GS Core Network and srsRAN Project with Simulated RF Environment using ZMQ Libraries

AN INTERNSHIP REPORT

Submitted by

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End-to-End Simulated 5G Network/Test Bed

Using Open5GS Core, srsRAN,
and ZMQ-Based RF Simulation

- Batch : SALEM.
- Team members : MANIKANDAN N, VIGNESH S.
- Center : Government Engineering College.

Problem Statement

- Develop full 5G SA simulation.
 - Use Open5GS Core and srsRAN for gNB/UE.
 - Implement ZMQ-based simulated RF links.
 - Study effects of latency and packet loss.
 - Evaluate throughput and reliability.

Objectives

- Build virtual 5G testbed.
 - Enable controlled network impairments.
 - Perform quantitative performance analysis.
 - Ensure experiment reproducibility.
 - Evaluate behavior under varying conditions.

System Architecture

- Open5GS 5G Core (AMF, SMF, UPF, UDM, AUSF, NRF).
 - srsRAN gNB and UE modules.
 - ZMQ-based virtual RF interfaces.
 - iperf3 for traffic generation.
 - Wireshark and logs for monitoring.

Major Components Used

- Open5GS: 5G Core.
 - srsRAN: RAN simulation.
 - ZMQ: Virtual RF channels.
 - Linux VM/container environment.
 - Traffic and analysis tools.

Implementation Steps

- Install and configure Open5GS.
 - Configure srsRAN for ZMQ RF interface.
 - Connect gNB with the 5G Core network.
 - Validate UE registration and PDU session.
 - Inject latency and packet loss.
 - Measure throughput and performance.

ZMQ-Based RF Simulation

- Virtual radio interface replaces hardware.
 - Supports controlled impairments.
 - Provides reproducible testing conditions.
 - Zero dependency on RF front-end.
 - Ideal for lab research and prototyping.

Performance Metrics

- Downlink/Uplink throughput.
 - Latency and jitter.
 - Packet loss rates.
 - Session/registration success rates.
 - System stability under impairments.

Expected Outcomes

- Fully functional 5G SA simulation.
 - Ability to test degradation scenarios.
 - Insights into latency-sensitive behavior.
 - Reliability and QoS interpretation.
 - Research-ready evaluation platform.

Applications & Future Work

- Academic 5G/6G research.
 - Testing scheduling and slicing algorithms.
 - Security and penetration testing.
 - 6G experimental prototyping.
 - Multi-cell and mobility extensions.

Nov 17 15:44

user@testbed:~

5GTech: Intern x 5G Simulation: X (91) WhatsApp x OFDM transmi x + -

http://localhost:3000/d/525a94d8-e16f-4b62-b904-247f08 Sign in

Home > Dashboards > 5G Simulation Dashboard

Last 5 minutes Refresh

UE Throughput (Mbps)

15:35

0-20

gNB Throughput (Mbps)

20

15:35

0-20

Packet Loss (%)

15:35

0-80

Latency (ms)

15:35

0-100

Open5GS CPU Usage ...

15:35

0-100

Open5GS Memory Us...

15:35

0-100

user@testbed:~

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5G Simulation Full Setup Complete
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user@testbed: $
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